BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

Section 'A' (Multiple Choice Questions) (17)

Maximum Marks: 17

Note: This section consists of 17 part questions and all are to be answered. Each question carries one mark. Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part. iii) The use of scientific calculator is allowed. (Options in the original are separated by star marks, shown here as A–D in printed order.)

Note: Time: 3 Hours — Maximum Marks: 85 (Science Pre-Engineering & Pre-Medical Groups)

Q1. Choose the correct answer for each from the given options:

Qi. This give violet colour to the flame:

(A) Li
(B) Na
(C) K
(D) Be

Qii. The order of reducing power of halide ion is:

(A) $I^- > Br^- > Cl^- > F^-$
(B) $F^- > Cl^- > Br^- > I^-$
(C) $I^- > Cl^- > F^- > Br^-$
(D) $Br^- > Cl^- > I^- > F^-$

Qiii. The coordination number of $[Cr(en)_3]^{+3}$ is:

(A) 3
(B) 4
(C) 5
(D) 6

Qiv. This fertilizer provide Nitrogen and Phosphorus in the plants:

(A) Urea
(B) Potassium Nitrate
(C) Ammonium Nitrate
(D) Diammonium Phosphate

Qv. This is hottest part of atmospher.:

(A) Thermosphere
(B) Mesosphere
(C) Stratosphere
(D) Troposphere

Qvi. Cathode in Castner Kellner Cell is:

(A) Titanium Blocks
(B) Carbon rods
(C) Iron Container
(D) Mercury

Qvii. Diagonal relationship of lithium with:

(A) Be
(B) Na
(C) Ca
(D) Mg

Qviii. This is low quality coal:

(A) Bituminous
(B) Lignite
(C) Anthracite
(D) Peat

Qix. This is used as an Ink preservative:

(A) Benzene
(B) Toluene
(C) Phenol
(D) Acetophenone

Qx. This one is more acidic:

(A) Carboxylic acid
(B) Phenol
(C) Alcohol
(D) Water

Qxi. This is meta directing group:

(A) $-OH$
(B) $-NH_2$
(C) $-CN$
(D) $-OR$

Qxii. Lucas reagent is a mixture of:

(A) Zn and Hg
(B) Zn and HCl
(C) $ZnCl_2$ and HCl
(D) NaOH and CaO

Qxiii. The hybridization in Carbonyl Carbon of aldehydes and ketones is:

(A) sp
(B) sp^2
(C) sp^3
(D) dsp^3

Qxiv. The molecule contains Ketoic group:

(A) Glucose
(B) Galactose
(C) Mannose
(D) Fructose

Qxv. Ozone layer is present in:

(A) Troposphere
(B) Stratosphere
(C) Mesosphere.
(D) Thermosphere

Qxvi. Carbolic acid contains this functional group:

(A) $-COOH$ (B) $-NO_2$

(C) $-OH$ (D) $-SO_3H$

Qxvii. In s-block element this from superoxide:

(A) Mg (B) Li

(C) Na (D) Cs

Section 'B' (Short-Answer Questions) (36)

Maximum Marks: 36

> Note: Answer any NINE part questions. Select FOUR part questions from Inorganic-General Chemistry and FIVE part questions from Organic Chemistry. All part questions carry equal marks.

QINORGANIC-GENERAL CHEMISTRY.

Q2.

(i) Discuss the group trend of ionization energy and electronegativity in group IA, IIA and IIIA.

OR Give any four scientific reasons of the following: * Alkaline earth metal possess smaller atomic radii and Alkali metal. * Acidity of hydrogen halides increases from HF to HI. * Ionization energy decreases from top to bottom in s-block elements. * Ligands are generally called Lewis-Bases. * Chromium exist in $4s^1,3d^5$ configuration but not in $4s^2,3d^4$.

(ii) Write the IUPAC name of the following: *$[Cr(en)_3](NO_3)_3$ *$[Pt(NH_3)_2Cl_2]^{2+}$ *$[Zn(OH)_4]^{2-}$ *$Na_3[Co(C_2O_4)_3]$

(iii) Explain any two of the following: *Catalytic Converter *CFC *Industrial Smog

(iv) What is Bleaching Powder? Write its Preparation and properties.

(v) Explain any two of the following properties of 3d series of transition elements: *Colour of Complexes *Catalytic Activity *Binding Energy

(vi) Explain the auto-oxidizing and reducing properties of chlorine and give classification of Ligands.

(vii) Define any four of the following: *Analgesic *Antibiotic *Chiral Carbon *Antipyretic Medicine *Anti-inflammatory Medicine.

—OR— What is diagonal relationship? Explain the diagonal relation between: *Li and Mg *Be and Al

QORGANIC CHEMISTRY.

(viii) Define any four of the following: *Catenation *Bucky Ball *Functional Group *Isomerism *Reforming of Petroleum

—OR— Describe homologous series of organic compound and write three properties of homologous series.

(ix) Write any four IUPAC name of the following: *$CH_3(CH_2)_5COO\ C(CH_3)_3$ *$(CH_2)_5(COOH)_2$. * (structure: benzene ring with −OH, −Br and −Cl substituents) * $CH_3 - CH_2 - OC(CH_3)_2C_2H_5$ * $CH_3 - CH_2 - C(=O) - CH_2 - COOH$

(x) Draw and explain orbital structure of Ethyne and write two methods of its preparation.

—OR— What are amines? Give three methods of preparation of amines.

(xi) Write equation and mechanism of sulphonation and nitration of benzene.

(xii) What is Free Radical? Explain the reaction mechanism of chlorination of methane in the presence of sunlight.

(xiii) Write the chemical equation of any four of the following: *Reaction of Ethyne with HBr. *Reaction of ethanol with acidified $K_2Cr_2O_7$. *Reaction of Grignard reagent with CO_2 *Reaction of ethandiol with HIO_4 *Reaction of ethanal with alcohol.

(xiv) Classify various proteins on the basis of functions and write its importance.

—OR— Explain the role of glucose, sucrose and lactose in health and disease.

Section "C" (Detailed Answer Questions) (32)

Maximum Marks: 32

> Note: Answer any Two questions – One question from Inorganic – General Chemistry and One from Organic Chemistry. Both questions carry equal marks.

QINORGANIC-GENERAL CHEMISTRY.

Q3.

(a) Describe the manufacturing of H_2SO_4 by contact process. Show by equation that is an oxidizing agent and dehydrating agent.

—OR— Describe the manufacture of Caustic soda by Castner-kellne cell. What are the advantages and disadvantages of the process?

(b) Write any Eight chemical equation for the following: * A mixture of nitrogen and silicon is heated under elevated temperature. * Phosphorous is put into water. * Chlorine gas is passed through hot solution of Caustic soda. * Reaction between $KMnO_4$ and $FeSO_4$ (write ionic equation). * Reaction of Mn with air. * Reaction of copper with concentrated HNO_3 * Reaction of Aluminum with caustic soda. * Reaction of Rubidium with air. * Reaction of Fluorine with oxygen. * Reaction of chromium with water.

Q4.

(a) Describe the various steps involve in the extraction of 99.99% pure copper from its chalcopyrite ore. Write the composition of duralumin and brass.

(b) Write notes on any three of the following *PVC *Nylon *Fertilizer *Detergent

QORGANIC CHEMISTRY.

Q5.

(a) What is meant by nucleophile? Give mechanism of the following reacti ns: * SN^1 reaction between 3° alkyl halide and NaCN. * SN^2 reaction between 1° alkyl halide and NaOH. Write three methods for the preparation of Alcohol.

(b) Draw the structure of any eight of the following: *Benzene –1, 4 – dioic acid *ethyl neopentyl ether *Benzophenone *p-cresol Di-vinyl acetylene * Neo pentyl alcohol * β − methyl butyric acid * 3 – chlorobenzamide * Resorcinol * tert, butyl chloride

Q6.

(a) Explain molecular orbital treatment of benzene and give the preparation of TNT, m-nitrotoluene and cyclohexane.

—OR— What are organometallic compounds? How is Grignard reagent prepared? Write the reactions of CH_3MgI with water, alcohol, ester and amines.

Q6.

(b) Write any eight chemical equations of the following: * Reaction of acetaldehyde with Zn/Hg amalgam. *Reaction of ethane with ozone. *Reaction of Grignard reagent with acetone. *Reaction of benzene diazonium chloride when heated with water. *Reaction of acetic acid with $LiAlH_4$. *Reaction of ethanol with acetic acid in the presence of H_2SO_4. * Reaction of bromoethane with KSH. * Reaction of phenol with concentrated HNO_3. *Reaction of alcohol with concentrated H_2SO_4. *Reaction of ethyne with water under specific condition.